Walmart Data Exploration

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Walmart Temperature and Weekly Sales

The following code loads the dataset and constructs the variables required for the empirical analysis of whether higher temperatures causally affect weekly sales.

```
# Load required packages
library(tidyverse)

Warning: package 'tidyverse' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0
──
✓ dplyr     1.1.4     ✓ readr     2.1.5
✓ forcats   1.0.1     ✓ stringr   1.5.2
✓ ggplot2   4.0.0     ✓ tibble    3.3.0
✓ lubridate 1.9.5     ✓ tidyr     1.3.1
✓ purrr     1.1.0
── Conflicts ────────────────────────────────────────── tidyverse_conflicts()
──
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors

library(lubridate)
library(fixest)

Warning: package 'fixest' was built under R version 4.5.2

library(modelsummary)

Warning: package 'modelsummary' was built under R version 4.5.2
```

This code imports the Walmart weekly sales dataset into R.

```
df <- read_csv("Walmart_Sales.csv")

Rows: 6435 Columns: 8
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr (1): Date
dbl (7): Store, Weekly_Sales, Holiday_Flag, Temperature, Fuel_Price, CPI,
Un...
```

```
ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

Constructing Variables

The following code converts dates into usable formats and constructs the log of weekly sales to allow percentage-based interpretation.

```r
df <- df %>%
  mutate(
    Date = parse_date_time(Date, orders = c("dmy","mdy","dmY","mdY")),
    Date = as.Date(Date),
    Store = factor(Store),
    year  = year(Date),
    month = month(Date),
    # log transform for interpretability and to reduce skew
    log_sales = log(Weekly_Sales)
  )

# quick checks
stopifnot(sum(is.na(df$Date)) == 0)
summary(select(df, Weekly_Sales, Temperature, Fuel_Price, CPI, Unemployment))
```

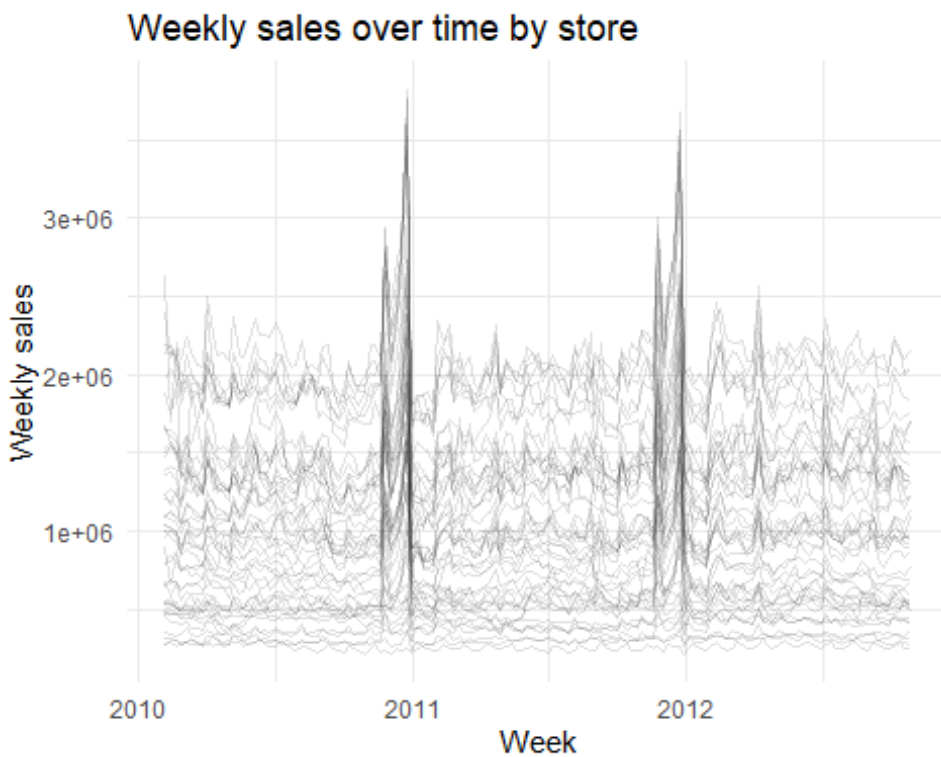
```
  Weekly_Sales      Temperature       Fuel_Price         CPI
 Min.   : 209986   Min.   : -2.06   Min.   :2.472   Min.   :126.1
 1st Qu.: 553350   1st Qu.: 47.46   1st Qu.:2.933   1st Qu.:131.7
 Median : 960746   Median : 62.67   Median :3.445   Median :182.6
 Mean   :1046965   Mean   : 60.66   Mean   :3.359   Mean   :171.6
 3rd Qu.:1420159   3rd Qu.: 74.94   3rd Qu.:3.735   3rd Qu.:212.7
 Max.   :3818687   Max.   :100.14   Max.   :4.468   Max.   :227.2
  Unemployment
 Min.   : 3.879
 1st Qu.: 6.891
 Median : 7.874
 Mean   : 7.999
 3rd Qu.: 8.622
 Max.   :14.313
```

Exploratory Analysis:

Sales over Time

This graph visualizes weekly sales patterns across stores and motivates the use of store and time fixed effects.

```
ggplot(df, aes(x = Date, y = Weekly_Sales, group = Store)) +
  geom_line(alpha = 0.12) +
  labs(x = "Week", y = "Weekly sales", title = "Weekly sales over time by
store") +
  theme_minimal()
```



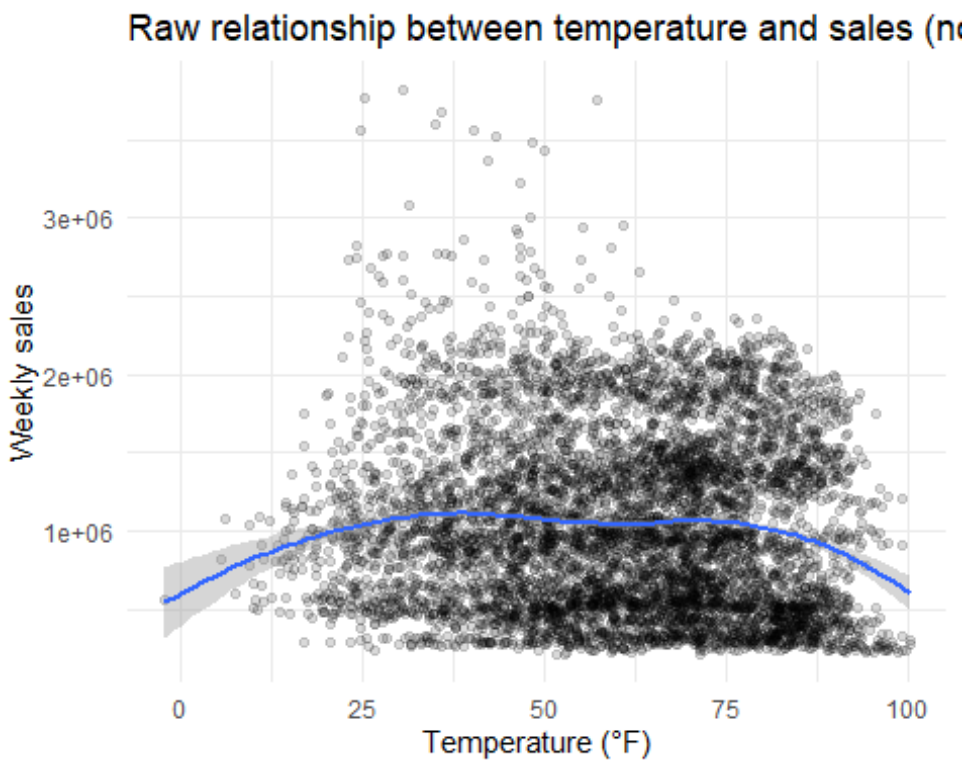
The persistent differences across stores and clear seasonal movements motivate the inclusion of store fixed effects and seasonal time controls in the regression analysis.

Temperature and Sales Relationship

The following plot explores the raw association between temperature and sales.

```
ggplot(df, aes(x = Temperature, y = Weekly_Sales)) +
  geom_point(alpha = 0.15) +
  geom_smooth(method = "loess") +
  labs(x = "Temperature (°F)", y = "Weekly sales",
       title = "Raw relationship between temperature and sales (not causal)")
+
  theme_minimal()

`geom_smooth()` using formula = 'y ~ x'
```



Because temperature is seasonal, this raw relationship does not identify a causal effect. However, it is useful for assessing potential nonlinear patterns, which are examined in later specifications.

Naive Regression

This model estimates the simple correlation between temperature and sales without controls.

```
m_naive <- feols(log_sales ~ Temperature, data = df)
etable(m_naive, digits = 3)

                           m_naive
Dependent Var.:          log_sales

Constant           13.9*** (0.025)
Temperature     -0.003*** (0.0004)
_______________ __________________
S.E. type                      IID
Observations                 6,435
R2                         0.00856
Adj. R2                    0.00841
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this specification, a one degree increase in temperature is associated with approximately a 0.3 percent decrease in weekly sales. However, this estimate does not control for store level heterogeneity or seasonal factors and therefore likely reflects omitted variable bias.

Fixed Effects Model

This specification controls for store and seasonal time effects to better isolate the causal effect of temperature.

```
m_main <- feols(
  log_sales ~ Temperature + Holiday_Flag + Fuel_Price + CPI + Unemployment |
    Store + year + month,
  data = df,
  vcov = ~Store
)

etable(m_naive, m_main, digits = 3)

                           m_naive            m_main
Dependent Var.:          log_sales         log_sales

Constant           13.9*** (0.025)
Temperature     -0.003*** (0.0004) 0.001*** (0.0003)
Holiday_Flag                        0.030*** (0.006)
Fuel_Price                           -0.015* (0.008)
CPI                                   0.0004 (0.005)
Unemployment                         -0.026* (0.012)
Fixed-Effects:  ------------------ -----------------
```

```
Store                           No                Yes
year                            No                Yes
month                           No                Yes
_______________ _________________ _________________
S.E. type                      IID          by: Store
Observations                 6,435              6,435
R2                         0.00856            0.96737
Within R2                       --            0.01633
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

In this specification, a one degree increase in temperature is associated with approximately a 0.10 percent increase in weekly sales, holding store characteristics and seasonal effects constant. A 10 degree increase corresponds to roughly a 1 percent increase in sales. The reversal in sign relative to the naive model indicates that failing to control for seasonal patterns biases the simple correlation.

Extracting and Comparing Temperature Coefficients

The following code extracts the estimated coefficient and clustered standard error for Temperature from the main fixed effects model. This allows us to directly report and interpret the magnitude and precision of the estimated effect. The code then estimates a quadratic specification to test whether the impact of temperature varies at higher levels and compares the results to the baseline model.

```
bT <- coef(m_main)[["Temperature"]]
seT <- se(m_main)[["Temperature"]]

bT

[1] 0.001035432

seT

[1] 0.0002887726

m_quad <- feols(
  log_sales ~ Temperature + I(Temperature^2) + Holiday_Flag + Fuel_Price +
CPI + Unemployment |
    Store + year + month,
  data = df,
  vcov = ~Store
)

etable(m_main, m_quad, digits = 3)
```

```
                                 m_main             m_quad
Dependent Var.:               log_sales          log_sales

Temperature           0.001*** (0.0003)  0.004*** (0.0008)
Holiday_Flag           0.030*** (0.006)   0.032*** (0.006)
Fuel_Price              -0.015* (0.008)    -0.019* (0.008)
CPI                      0.0004 (0.005)   -7.08e-5 (0.005)
Unemployment            -0.026* (0.012)    -0.028* (0.012)
I(I(Temperature^2))                     -3.01e-5*** (7e-6)
Fixed-Effects:        ---------------- ------------------
Store                              Yes                Yes
year                               Yes                Yes
month                              Yes                Yes
_________________ _________________ __________________
S.E.: Clustered               by: Store          by: Store
Observations                      6,435              6,435
R2                              0.96737            0.96765
Within R2                       0.01633            0.02468
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

This code extracts the point estimate and clustered standard error for Temperature from the fixed effects model to quantify the magnitude and statistical precision of the effect. It then estimates a quadratic specification to assess whether the relationship between temperature and sales is nonlinear and compares it to the baseline model.

Testing for Delayed Effects of Temperature

This code constructs a one-week lag of temperature within each store and re-estimates the fixed-effects model, including both current and lagged temperature, to test whether temperature affects sales with a delay.

```
df_lag <- df %>%
  arrange(Store, Date) %>%
  group_by(Store) %>%
  mutate(lag_temp = lag(Temperature, 1)) %>%
  ungroup()

m_lag <- feols(
  log_sales ~ Temperature + lag_temp + Holiday_Flag + Fuel_Price + CPI +
Unemployment |
    Store + year + month,
  data = df_lag,
  vcov = ~Store
)

NOTE: 45 observations removed because of NA values (RHS: 45).
```

```
etable(m_main, m_lag, digits = 3)

                           m_main             m_lag
Dependent Var.:         log_sales         log_sales

Temperature     0.001*** (0.0003) 0.001*** (0.0003)
Holiday_Flag     0.030*** (0.006)  0.032*** (0.006)
Fuel_Price        -0.015* (0.008)   -0.018* (0.008)
CPI                0.0004 (0.005)   3.22e-5 (0.005)
Unemployment      -0.026* (0.012)   -0.026* (0.012)
lag_temp                          -0.0005. (0.0003)
Fixed-Effects:  ---------------- ----------------
Store                        Yes               Yes
year                         Yes               Yes
month                        Yes               Yes
_______________ _________________ _________________
S.E.: Clustered         by: Store         by: Store
Observations                6,435             6,390
R2                        0.96737           0.96744
Within R2                 0.01633           0.01810
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Conclusion and Managerial Interpretation

The central question of this analysis was whether higher temperatures causally affect weekly sales across Walmart stores. While temperature and sales move together seasonally, the key issue is whether short run fluctuations in temperature affect sales after accounting for store differences and common seasonal patterns.

Using a fixed effects framework that compares each store to itself over time while controlling for month and year effects, I find that higher temperatures are associated with a small positive effect on weekly sales. Specifically, a 1°F increase in temperature is associated with approximately a 0.10 percent increase in weekly sales, holding store characteristics and seasonal factors constant. A 10°F increase corresponds to roughly a 1 percent increase in sales.

To put this into practical terms, for a store averaging one million dollars in weekly sales, a 10°F increase corresponds to roughly a ten thousand dollar increase in revenue. This is noticeable but not transformative relative to typical weekly fluctuations. Weekly sales often vary by substantially more than this due to routine demand cycles, promotional timing, and macroeconomic conditions. Although the estimated effect is statistically precise, its economic magnitude is modest.

The comparison between the naive and fixed effects models is particularly informative. Without controls, temperature appears negatively related to sales. Once store and time effects are included, the relationship becomes positive. This reversal indicates that simple correlations are confounded by seasonality and underlying demand patterns. Much of the apparent negative association reflects omitted seasonal factors rather than a true temperature effect.

Additional specifications confirm the stability of the main result. Allowing for nonlinear temperature effects suggests diminishing marginal impacts at very high temperatures, but the overall economic magnitude remains small. Similarly, including lagged temperature provides little evidence of meaningful delayed effects. Together, these robustness checks suggest that the core finding of a small contemporaneous positive effect is not sensitive to reasonable alternative specifications.

From a managerial standpoint, the results imply that short run temperature fluctuations are not a major driver of weekly sales performance. While warmer weeks are associated with slightly higher sales, the effect is too small to meaningfully alter revenue forecasts or operational planning decisions. Factors such as pricing strategy, promotional timing, inventory management, and broader economic conditions are likely to play substantially larger roles in determining sales outcomes.

Although the fixed effects approach strengthens causal credibility by comparing each store to itself over time, unobserved time varying local shocks may still influence the estimates. Therefore, the findings should be interpreted as credible but not definitive causal effects.

Overall, the evidence suggests that while temperature has a measurable impact on weekly sales, its magnitude is small. Within the range observed from 2010 to 2012, rising temperatures do not pose a meaningful short run risk or opportunity for weekly Walmart sales performance.